



8.2. Data Visualization: Statistical Plots

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While categorical plots compare values across groups, **statistical plots** focus on understanding the characteristics of numerical data. They help us examine distributions, identify outliers, explore relationships between variables, and measure correlations.

These visualizations play an important role in **Exploratory Data Analysis (EDA)**, allowing analysts to understand the structure of a dataset before applying statistical methods or machine learning algorithms.

Throughout this section, we continue using the synthetic student dataset introduced in the previous section.

```

import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

np.random.seed(42)

n = 200

df = pd.DataFrame({
    "Student ID": [f"S{i:03d}" for i in range(1, n+1)],
    "Age": np.random.randint(18, 24, n),
    "Height": np.random.normal(170, 8, n).round(1),
    "Weight": np.random.normal(65, 12, n).round(1),
    "Gender": np.random.choice(["Male", "Female"], n),
    "Major": np.random.choice(["AI", "IT", "CSC", "SE"],
n),
    "GPA": np.round(np.random.uniform(2.0, 4.0, n), 2)
})

df.head()

```

1. Scatter Plot

A **scatter plot** visualizes the relationship between two numerical variables by representing each observation as a point. It is commonly used to determine whether two variables exhibit a positive, negative, or no apparent correlation.

For example, taller students generally tend to weigh more, suggesting a positive relationship between **Height** and **Weight**.

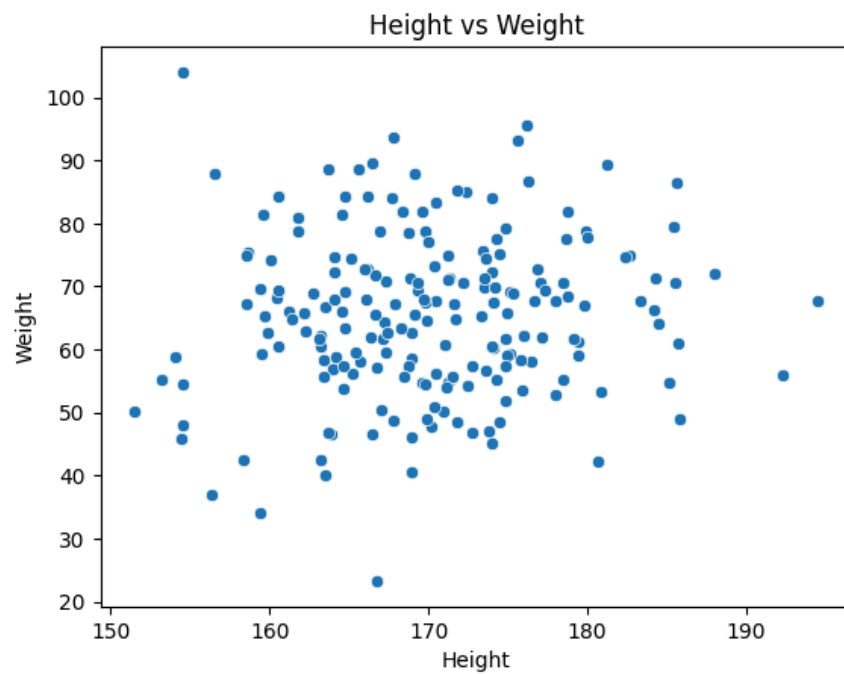
Example. Height vs Weight

```

sns.scatterplot(
    data=df,
    x="Height",
    y="Weight"
)

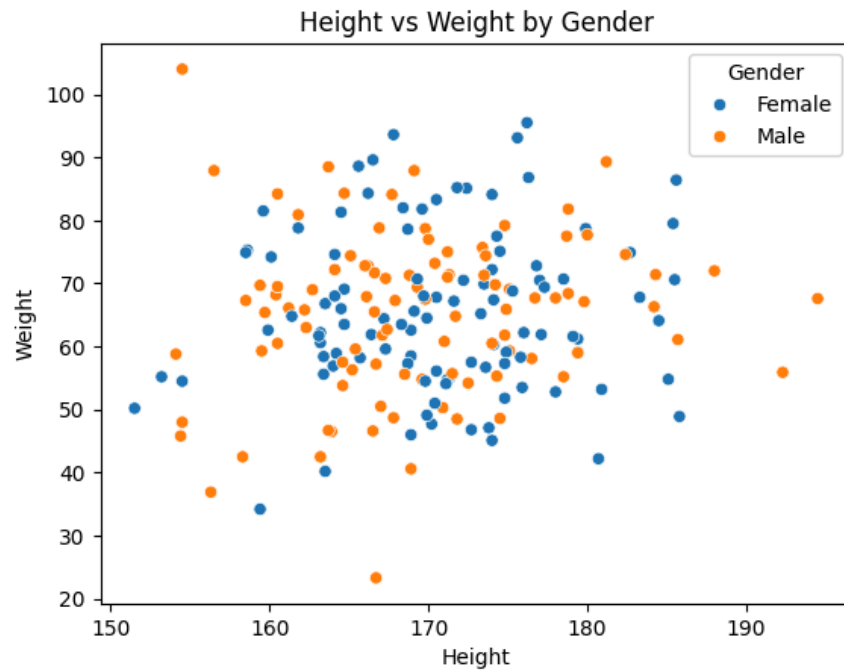
```

```
)  
  
plt.title("Height vs Weight")  
plt.show()
```



Example. Height vs Weight by Gender

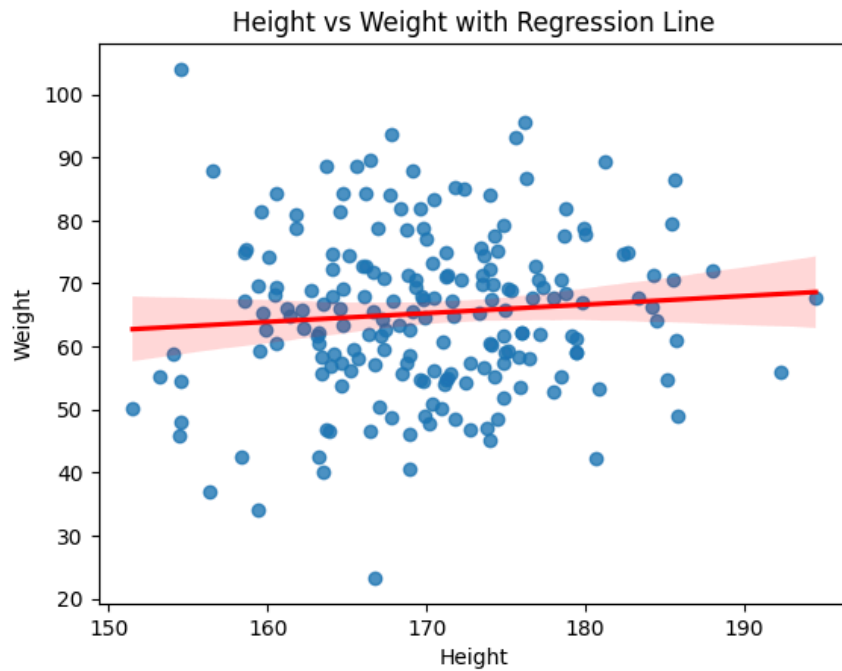
```
sns.scatterplot(  
    data=df,  
    x="Height",  
    y="Weight",  
    hue="Gender"  
)  
  
plt.title("Height vs Weight by Gender")  
plt.show()
```



Example. Scatter Plot with Regression Line

Adding a regression line helps visualize the overall linear relationship between the two variables.

```
sns.regplot(  
    data=df,  
    x="Height",  
    y="Weight",  
    line_kws={'color': 'red'}  
)  
  
plt.title("Height vs Weight with Regression Line")  
plt.show()
```



2. Box Plot

A **box plot** summarizes the distribution of a numerical variable using the five-number summary:

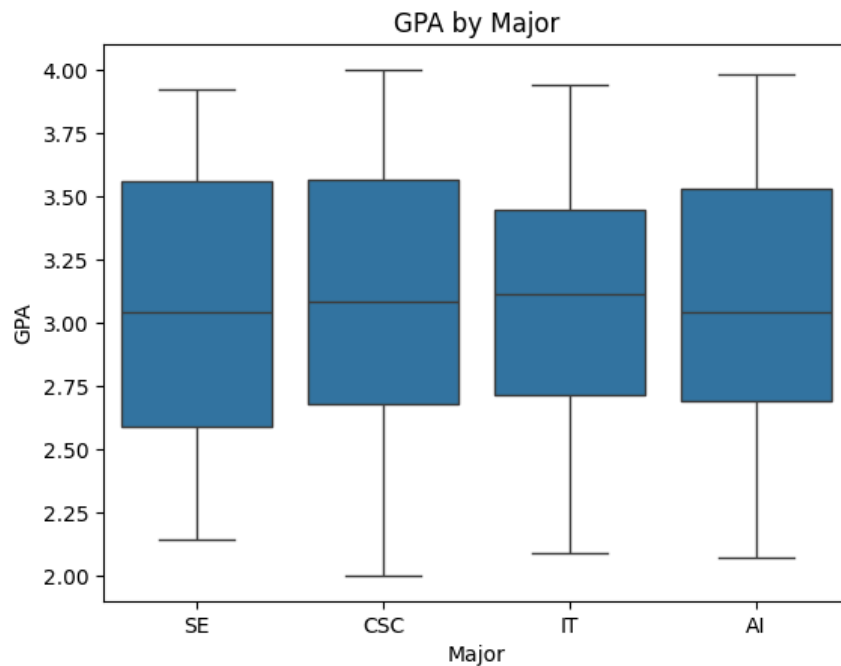
- Minimum (excluding outliers)
- First Quartile (Q1)
- Median
- Third Quartile (Q3)
- Maximum (excluding outliers)

The box represents the **Interquartile Range (IQR = Q3 – Q1)**, which contains the middle 50% of the data. Values outside the whiskers are displayed as **outliers**.

Example. GPA by Major

```
sns.boxplot(  
    data=df,  
    x="Major",  
    y="GPA"  
)
```

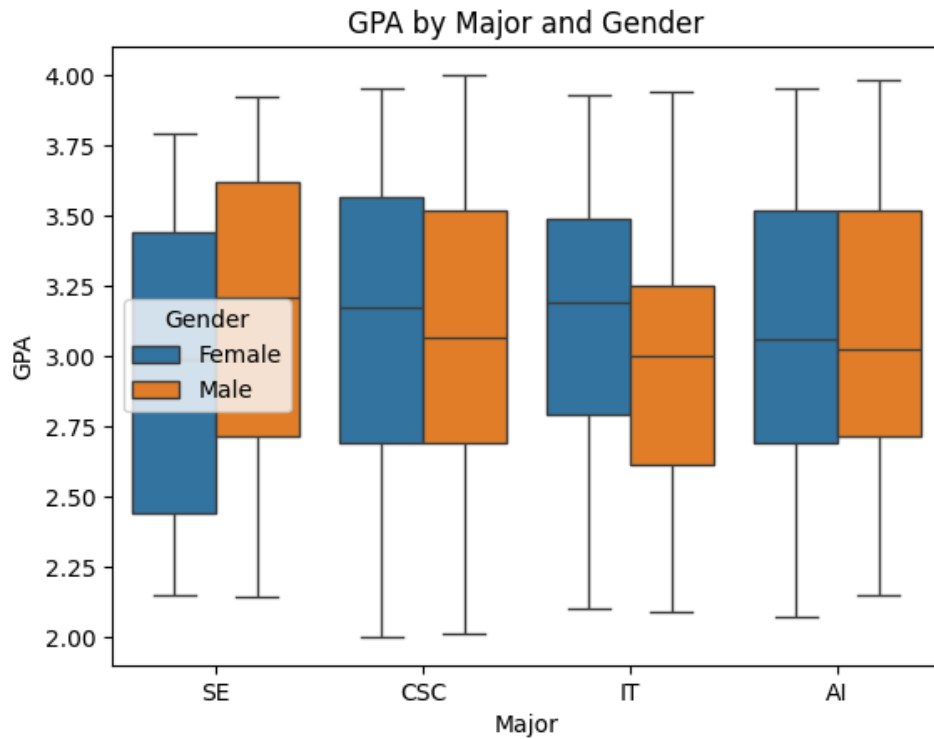
```
plt.title("GPA by Major")
plt.show()
```



Example. GPA by Major and Gender

```
sns.boxplot(
    data=df,
    x="Major",
    y="GPA",
    hue="Gender"
)

plt.title("GPA by Major and Gender")
plt.show()
```



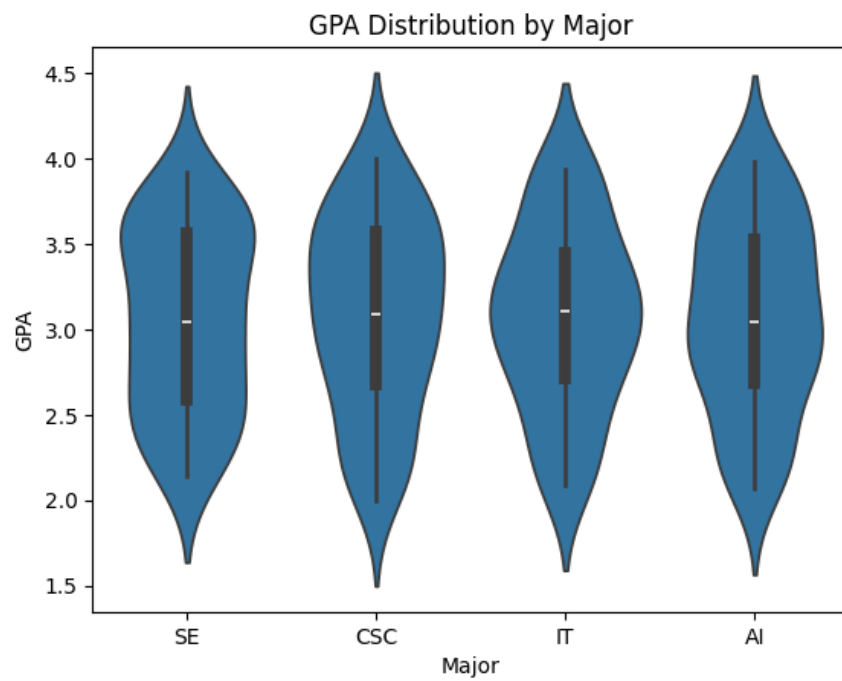
3. Violin Plot

A **violin plot** extends the box plot by displaying the full distribution of the data. In addition to the median and quartiles, it uses **Kernel Density Estimation (KDE)** to show where observations are concentrated.

Violin plots are particularly useful when comparing multiple distributions or identifying multimodal data.

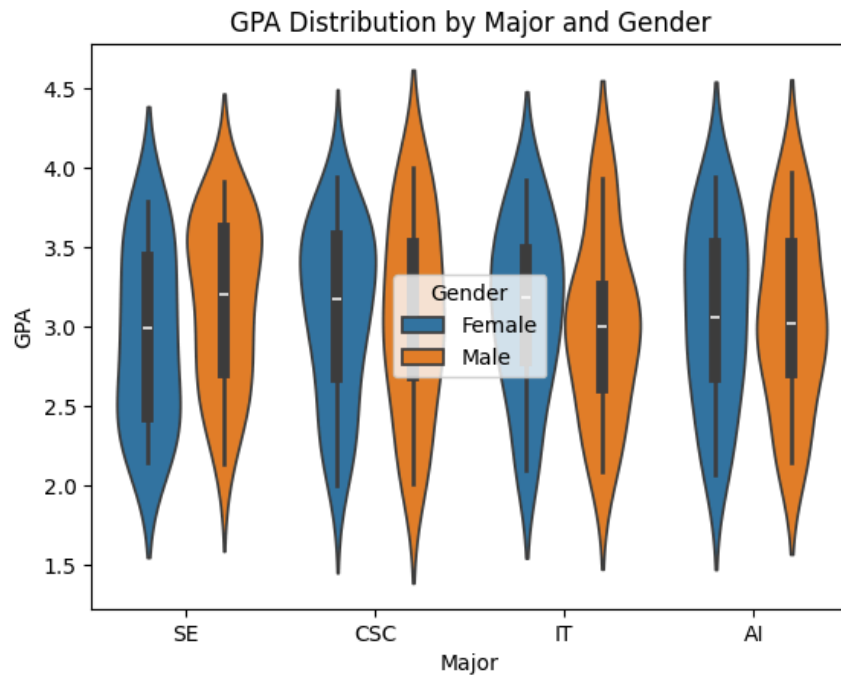
Example. GPA by Major

```
sns.violinplot(  
    data=df,  
    x="Major",  
    y="GPA"  
)  
  
plt.title("GPA Distribution by Major")  
plt.show()
```



Example. GPA by Major and Gender

```
sns.violinplot(  
    data=df,  
    x="Major",  
    y="GPA",  
    hue="Gender"  
)  
  
plt.title("GPA Distribution by Major and Gender")  
plt.show()
```

4. Histogram

A **histogram** displays the distribution of a continuous variable by grouping observations into intervals called **bins**.

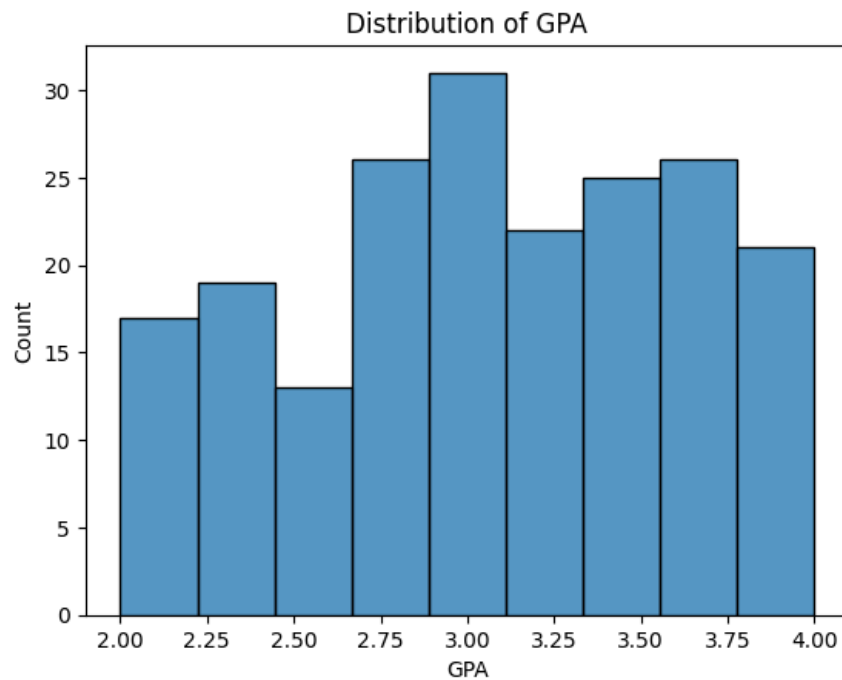
Histograms help identify characteristics such as:

- Central tendency
- Spread
- Skewness
- Number of peaks (modality)
- Approximate normality

Example. GPA Distribution

```
sns.histplot(  
    data=df,  
    x="GPA"  
)
```

```
plt.title("Distribution of GPA")
plt.show()
```

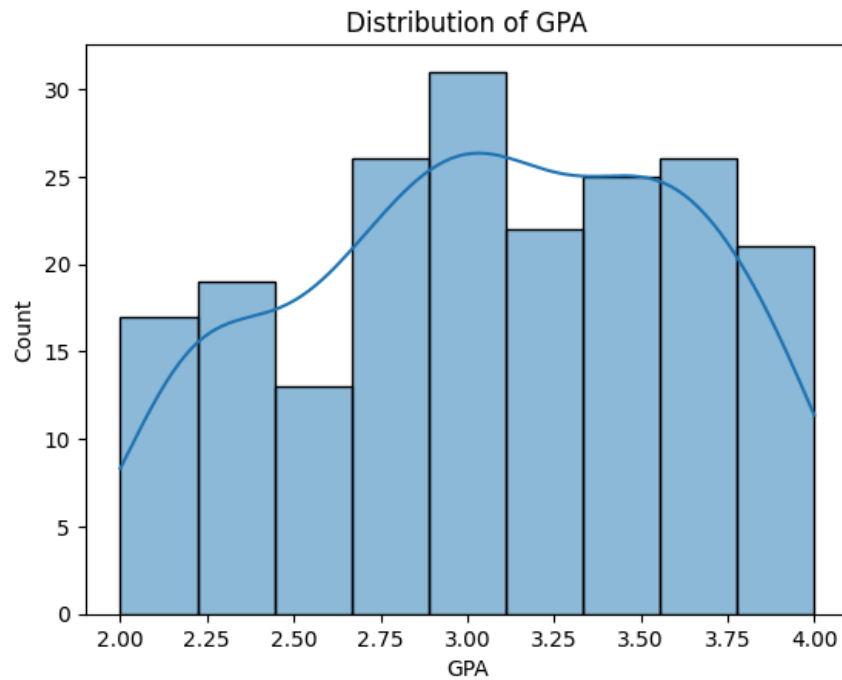


Example. Histogram with KDE

Adding a KDE curve provides a smoother approximation of the underlying distribution.

```
sns.histplot(
    data=df,
    x="GPA",
    kde=True
)

plt.title("Distribution of GPA")
plt.show()
```

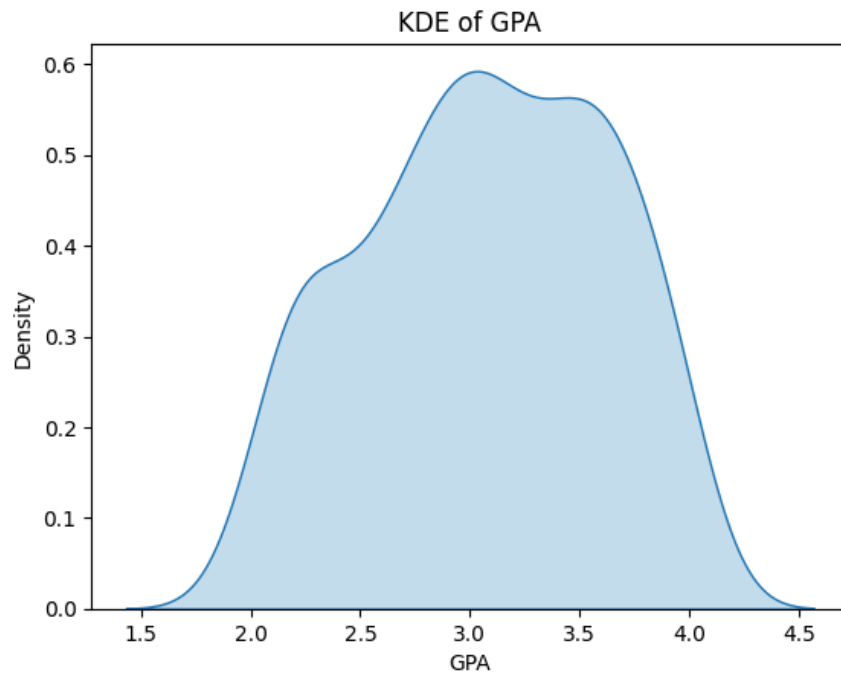


5. KDE Plot

A **Kernel Density Estimation (KDE) plot** estimates the probability density of a continuous variable using a smooth curve. Unlike histograms, KDE plots do not depend on bin sizes, making them especially useful for comparing multiple distributions.

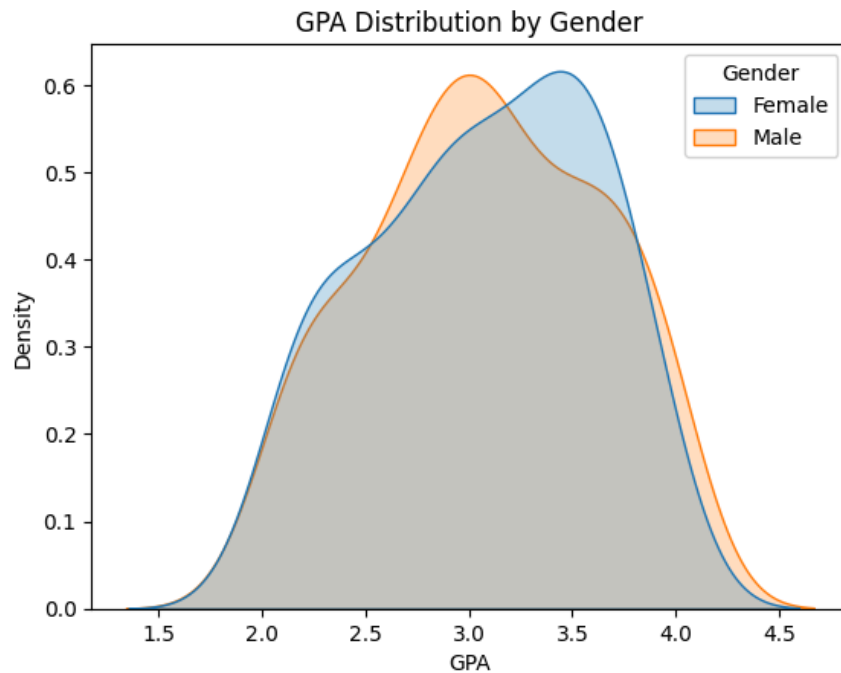
Example. GPA Distribution

```
sns.kdeplot(  
    data=df,  
    x="GPA",  
    fill=True  
)  
  
plt.title("KDE of GPA")  
plt.show()
```



Example. GPA Distribution by Gender

```
sns.kdeplot(  
    data=df,  
    x="GPA",  
    hue="Gender",  
    fill=True,  
    common_norm=False  
)  
  
plt.title("GPA Distribution by Gender")  
plt.show()
```



6. Heatmap

A **heatmap** represents numerical values using color intensity. One of its most common applications is visualizing the **correlation matrix**, where colors indicate the strength and direction of relationships between numerical variables.

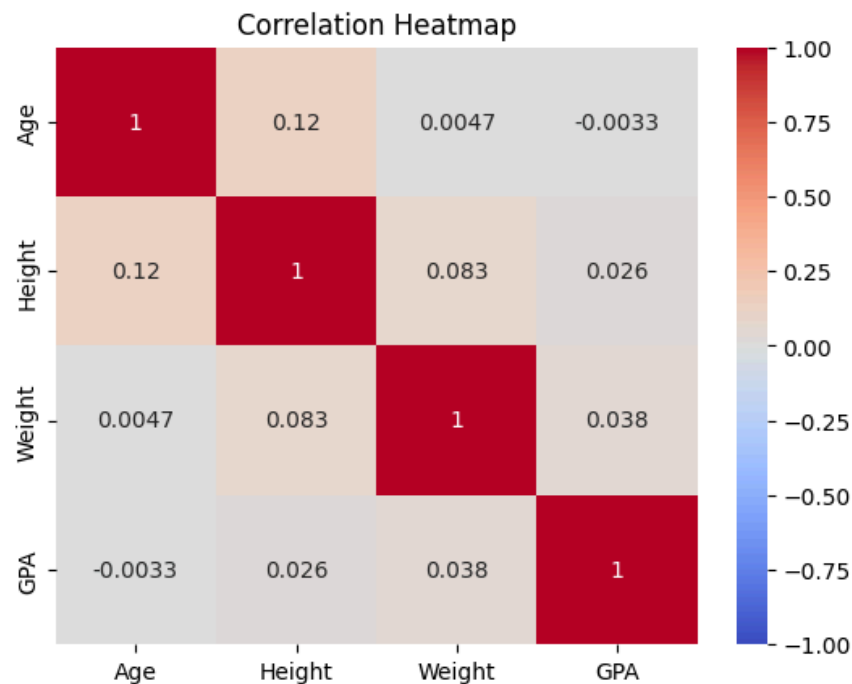
Correlation heatmaps are most appropriate for continuous numerical features and provide a quick overview of variables that are strongly related.

Example. Correlation Heatmap

```
corr = df.select_dtypes("number").corr()

sns.heatmap(
    corr,
    annot=True,
    cmap="coolwarm",
    vmin=-1,
    vmax=1
)
```

```
plt.title("Correlation Heatmap")
plt.show()
```



Summary

In this section, we explored the most widely used **statistical plots** for exploratory data analysis. Unlike categorical plots, these visualizations focus on understanding numerical variables by revealing their distributions, detecting outliers, identifying correlations, and examining relationships between features.

Together with the categorical plots from the previous section, these visualizations form the foundation of data exploration. Mastering them enables analysts to gain meaningful insights from data before applying statistical tests or building predictive machine learning models.

References

<https://seaborn.pydata.org/tutorial/relational.html>

<https://www.geeksforgeeks.org/data-analysis/types-of-data-visualization/>

Next Section:

 [9. Exploratory Data Analysis](#)